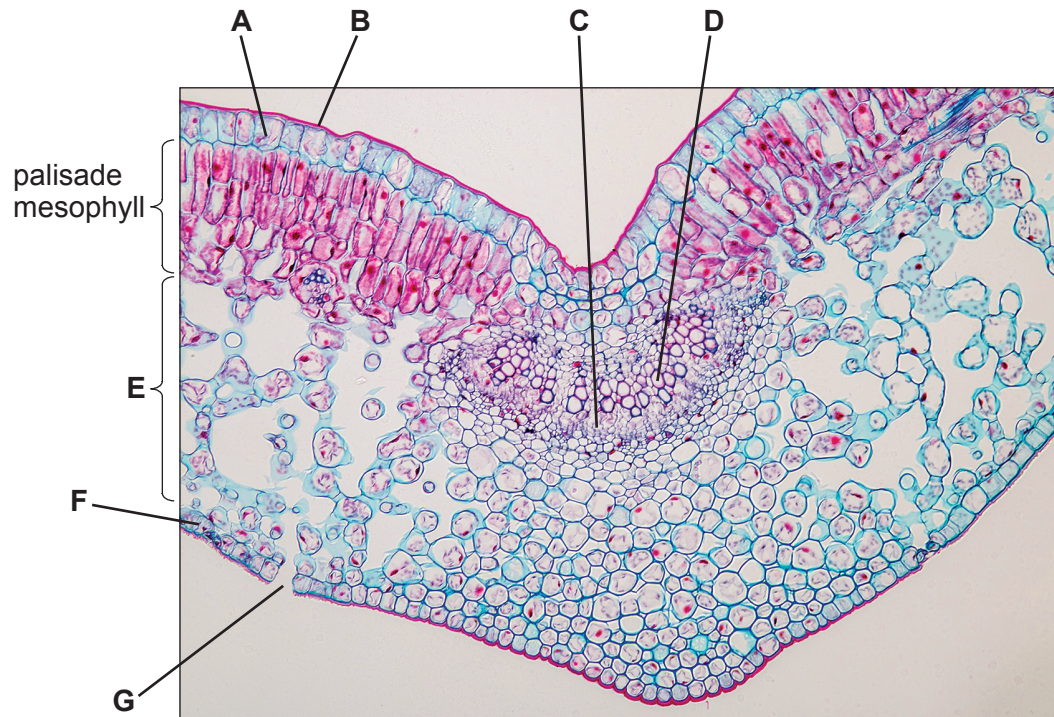


2. Leaves show many adaptations for photosynthesis. **Image 2.1** shows a transverse section through a leaf.

Image 2.1



- (a) The palisade mesophyll tissue, shown in **Image 2.1**, is the main site of photosynthesis. Identify and explain **two** adaptations of this **tissue** for photosynthesis. [2]

Adaptation 1

.....

.....

Adaptation 2

.....

.....



- (b) The chemical equation for photosynthesis is shown below:



Use **letters (A–G)** from **Image 2.1** to identify the following:

- (i) **Two** structures that allow the **reactants** to reach the palisade mesophyll cells. [1]

.....

- (ii) **Two** structures that allow the **products** to be moved from the palisade mesophyll cells. [1]

.....

- (c) A student concluded that **Image 2.1** was taken from a leaf of a mesophyte.
State **one** piece of evidence from **Image 2.1** that shows it was **not** taken from a:

- (i) hydrophyte; [1]

.....
.....

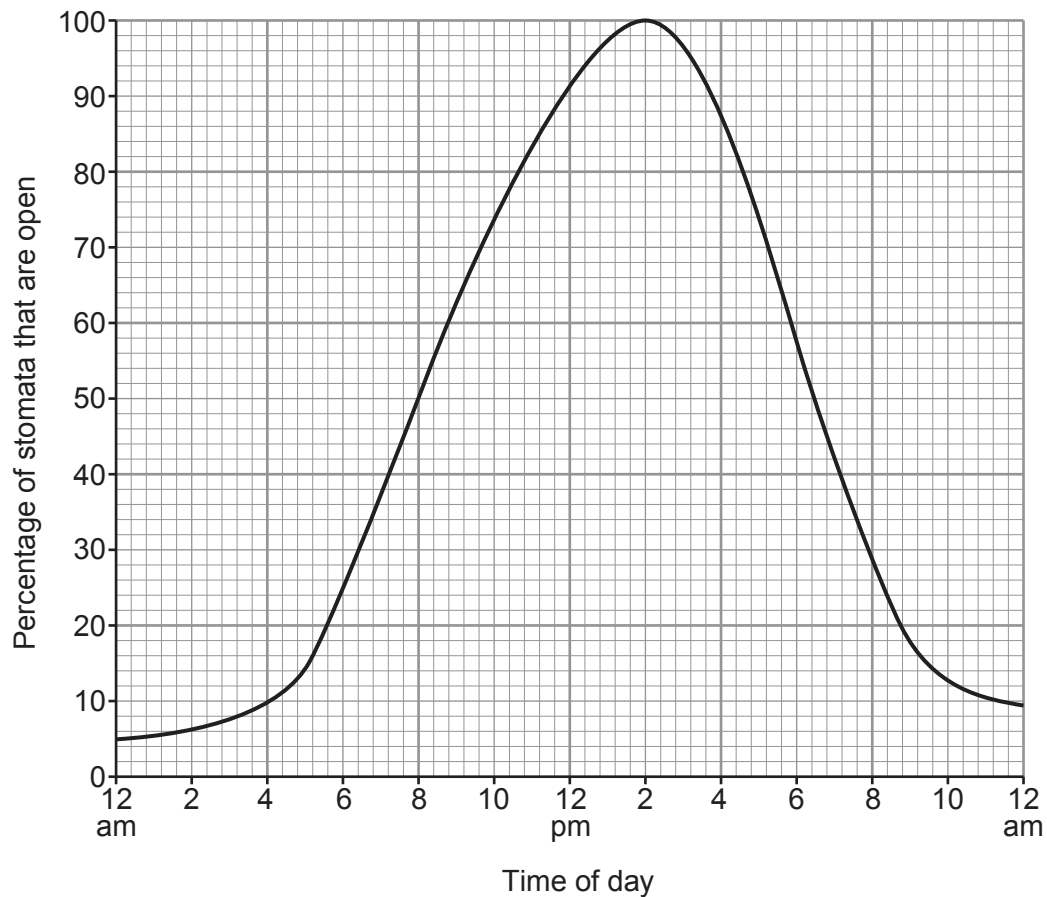
- (ii) xerophyte. [1]

.....
.....



- (d) **Graph 2.2** shows the percentage of stomata that are open at different times over a 24-hour period.

Graph 2.2



Describe the pattern of data shown in **Graph 2.2**. Explain the advantage of this to the plant. [4]

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